

MATHEMATICAL PENDULUM

An idealised mathematical pendulum consists of point mass suspended from a massless string. When the mass is deflected from its equilibrium position, it describes an oscillatory motion with a (constant) period.

We investigate how the period of a mathematical pendulum depends on the length of the string with the goal to suggest and verify a mathematical model.

MEASUREMENTS

SETUP

OSCILLATION PERIOD FOR DIFFERENT LENGTHS AND SMALL AMPLITUDE

The length ℓ corresponds to the distance between the pivot point and the centre of mass of the pendulum bob. For a given length, we measure the time Δt for a N periods. For a good accuracy, the measured time should be at least 10 s. To reduce the likelihood of human errors, we measure the time for each length at least twice. The period T can then be calculated as

$$T = \frac{\overline{\Delta t}}{N}$$

ℓ (cm)											$\pm$
N											
Δt_1 (s)											$\pm$
Δt_2 (s)											$\pm$
T (s)											

ANALYSIS

Using the provided Jupyter notebook as a starting point, analyse your data according to the following steps:

- Graph the period vs. length in an error plot (i.e. with error bars). Make an educated guess about the functional relation between the two quantities.
- Determine the fit parameter(s) (including uncertainty) for a possible fit function and add the graph of the fit function to the error plot. You may want to compare different options for the fit function.
- Calculate the residuals and add a residual plot to verify the quality of your preferred model.

ADDITIONAL TASKS

1. Determine the period for several pendula with the same length, but different masses. Theory predicts that the mass does not affect the period. Verify this hypothesis with your data.
2. Investigate the period for larger amplitudes. Determine the minimum amplitude where the measured period is significantly larger than the theoretical value.
3. The theoretical prediction for the period of a mathematical pendulum (for small amplitudes) is

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

Using the fit parameter for the square root fit, determine the value for g (with uncertainty).

4. Collect the data of all lab groups and repeat the analysis for the period as a function of pendulum length. Discuss how the larger number of data points affects the quality of the calculated results.